22 ATOMS AND RADIOACTIVITY

Atoms are made up of sub-atomic particles called neutrons, protons and electrons. It is the numbers of these particles that give each element its unique properties. In this chapter you will find out how atoms can break up and become transformed into different elements, and about the different types of radiation they give out.

LEARNING OBJECTIVES

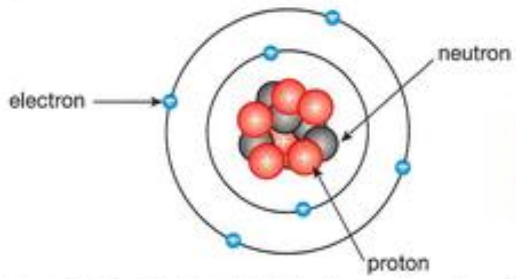
- Describe the structure of an atom in terms of protons, neutrons and electrons and use symbols such as $^{14}_{6}C$ to describe particular nuclei
- Know the terms atomic (proton) number, mass (nucleon) number and isotope
- Know that alpha (α) particles, beta (β^-) particles, and gamma (γ) rays are ionising radiations emitted from unstable nuclei in a random process
- Describe the nature of alpha (α) particles, beta (β^-) particles, and gamma (γ) rays, and recall that they may be distinguished in terms of penetrating power and ability to ionise
- Investigate the penetration powers of different types of radiation using either radioactive sources or simulations
- Describe the effects on the atomic and mass numbers of a nucleus of the emission of each of the four main types of radiation (alpha, beta, gamma and neutron radiation)
- Understand how to balance nuclear equations in terms of mass and charge

UNITS

In this section you will need to use becquerel (Bq) as the unit of activity of a radioactive source, centimetre (cm) as the unit of length, and minute (min) and second (s) as the units of time.

ELECTRONS, PROTONS AND NEUTRONS

Atoms are made up of electrons, protons and neutrons. Figure 22.1 shows a simple model of how these particles are arranged.



not to scale
The nucleus, made up of neutrons and protons, is about 10 000 times smaller than the atom itself.

▲ Figure 22.1 A simple model with protons and neutrons in the nucleus of the atom and electrons in orbits around the outside

The electron is a very small particle with very little mass. It has a negative electric charge. Electrons orbit the nucleus of the atom. The nucleus is very small compared to the size of the atom itself. The diameter of the nucleus is about 10 000 times smaller than the diameter of the atom. If the nucleus of an atom were enlarged to the size of a full stop on this page, the atom would have a diameter of around 2.5 metres.

The nucleus is made up of protons and neutrons. Protons and neutrons have almost exactly the same mass. Protons and neutrons are nearly 2000 times heavier than electrons. Protons carry positive electric charge but neutrons, as the name suggests, are electrically neutral or uncharged. The amount of charge on a proton is equal to that on an electron but opposite in sign.

EXTENSION WORK

The actual mass of an electron is 9.1×10^{-31} kg and its actual charge is -1.6×10^{-19} C. The table shows the approximate relative masses of the particles and the relative amount of charge they carry. The mass in atomic mass units is discussed below.

The properties of these three atomic particles are summarised in the table below. Protons and neutrons are also called nucleons because they are found in the nucleus of the atom.

Atomic particle	Relative mass of particle	Relative charge of particle
electron	1	−1
proton	2000	+1
neutron	2000	0

THE ATOM

The nucleus of an atom is surrounded by electrons. We sometimes think of electrons as orbiting the nucleus in a way similar to the planets orbiting the Sun. It is more accurate to think of the electrons as moving rapidly around the nucleus in a cloud or shell.

An atom is electrically neutral. This is because the number of positive charges carried by the protons in its nucleus is balanced by the number of negative charges on the electrons in the electron 'cloud' around the nucleus.

ATOMIC NUMBER, Z

The chemical behaviour and properties of a particular element depend upon how the atoms combine with other atoms. This is determined by the number of electrons in the atom. Although atoms may gain or lose electrons, sometimes quite easily, the number of protons in atoms of a particular element is always the same. The atomic number of an element tells us how many protons each of its atoms contains. For example, carbon has six protons in its nucleus – the atomic number of carbon is, therefore, 6. The symbol we use for atomic number is Z. Each element has its own unique atomic number. The atomic number is sometimes called the proton number.

ATOMIC MASS, A

The total number of protons and neutrons in the nucleus of an atom determines its atomic mass. The mass of the electrons that make up an atom is tiny and can usually be ignored. The mass of a proton is approximately 1.7×10^{-27} kg. To save writing this down we usually refer to the mass of an atom by its mass number or nucleon number. This number is the total number of protons and neutrons in the atom. The mass number of an element is given the symbol A.

ATOMIC NOTATION – THE RECIPE FOR AN ATOM

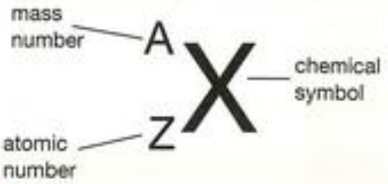
Each particular type of atom will have its own atomic number, which identifies the element, and a mass number that depends on the total number of nucleons, or particles, in the nucleus. Figure 22.2 shows the way we represent an atom of an element whose chemical symbol is X, showing the atomic number and the mass number.

So, using this notation, an atom of oxygen is represented by:

$$^{16}_{8}\mathrm{O}$$

The chemical symbol for oxygen is O. The atomic number is 8 – this tells us that the nucleus contains eight protons. The mass number is 16, so there are 16 nucleons (protons and neutrons) in the nucleus. Since eight of these are protons, the remaining eight must be neutrons. The atom is electrically neutral overall, so the +8 charge of the nucleus is balanced by the eight orbiting electrons, each with charge −1.

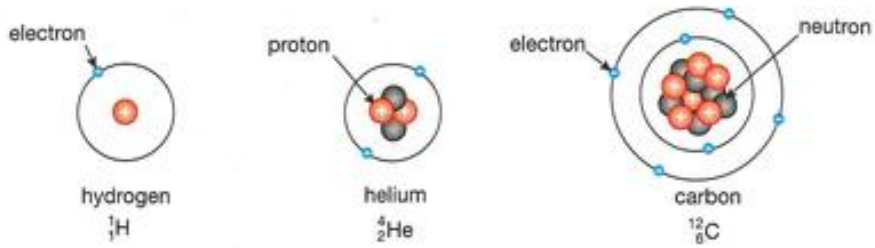
KEY POINT



▲ Figure 22.2 Atomic notation

mass number, A = number of neutrons + number of protons, Z
= number of nucleons

so
number of neutrons = number of nucleons – number of protons
= A – Z



▲ Figure 22.3 The hydrogen atom has one proton in its nucleus and no neutrons, so the mass number A = 1 + 0 = 1. As it has one proton, its atomic number, Z = 1. For helium, A = 4 (2 protons + 2 neutrons) and Z = 2 (2 protons). For carbon, A = 12 (6 protons + 6 neutrons) while Z = 6 (6 protons).

Figure 22.3 shows some examples of the use of this notation for hydrogen, helium and carbon, together with a simple indication of the structure of an atom of each of these elements. In each case the number of orbiting electrons is equal to the number of protons in the nucleus, so the atoms are electrically neutral.

ISOTOPES

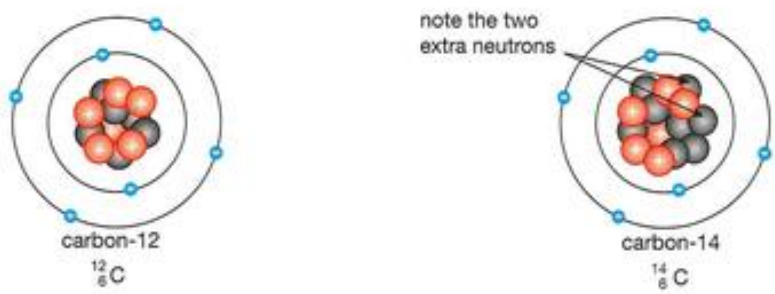
The number of protons in an atom identifies the element. The chemical behaviour of an element depends on the number of electrons it has and, as we have seen, this always balances the number of protons in the nucleus. However, the number of neutrons in the nucleus can vary slightly. Atoms of an element with different numbers of neutrons are called isotopes of the element. The number of neutrons in a nucleus affects the mass of the atom. Different isotopes of an element will all have the same atomic number, Z, but different mass numbers, A. Figures 22.4 and 22.5 show some examples of isotopes.



▲ Figure 22.4 Isotopes of hydrogen – they all have the same atomic number, 1, and the same chemical symbol, H.

EXTENSION WORK

Hydrogen-2 is also called heavy hydrogen or deuterium. Hydrogen-3 is called tritium.



▲ Figure 22.5 Two isotopes of carbon – they are referred to as carbon-12 and carbon-14 to distinguish between them.

THE STABILITY OF ISOTOPES

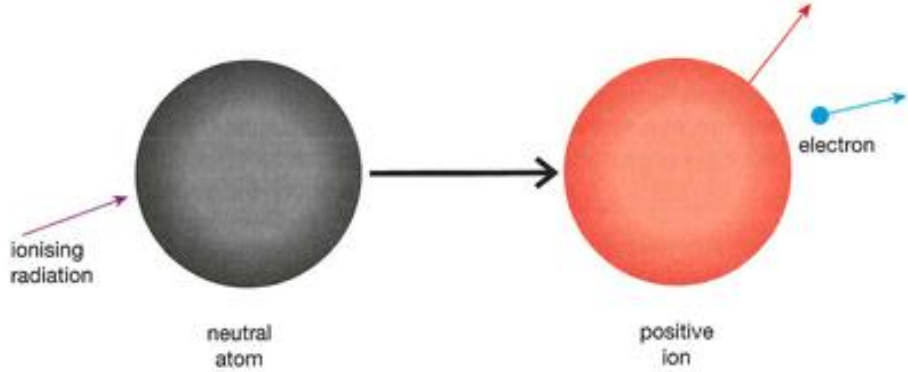
Isotopes of an element have different physical properties from other isotopes of the same element. One obvious difference is the mass. Another difference is the stability of the nucleus.

The protons are held in the nucleus by the nuclear force. This force is very strong and acts over a very small distance. It is strong enough to hold the nucleus together against the electric force repelling the protons away from

each other. (Remember that protons carry positive charge and like charges repel.) The presence of neutrons in the nucleus affects the balance between these forces. Too many or too few neutrons will make the nucleus unstable. An unstable nucleus will eventually decay. When the nucleus of an atom decays it gives out energy and may also give out alpha or beta particles.

IONISING RADIATION

When unstable nuclei decay they give out ionising radiation. Ionising radiation causes atoms to gain or lose electric charge, forming ions. Unstable nuclei decay at random. This means that it is not possible to predict which unstable nucleus in a piece of radioactive material will decay, or when decay will happen. We shall see that we can make measurements that will enable us to predict the probability that a certain proportion of a radioactive material will decay in a given time.



▲ Figure 22.6 When a neutral atom (or molecule) is hit by ionising radiation it loses an electron and becomes a positively charged ion.

There are three basic types of ionising radiation. They are: alpha (α), beta (β) and gamma (γ) radiation.

ALPHA (α) RADIATION



▲ Figure 22.7 An alpha particle (α particle)

EXTENSION WORK

Nucleons have roughly 2000 times the mass of an electron, and alpha particles are made up of four nucleons, so an alpha particle has 8000 times the mass of a beta particle.

Alpha radiation consists of fast-moving particles that are thrown out of unstable nuclei when they decay. These are called alpha particles. Alpha particles are helium nuclei – helium atoms without their orbiting electrons. Figure 22.7 shows an alpha particle and the notation that is used to denote it in equations.

Alpha particles have a relatively large mass. They are made up of four nucleons and so have a mass number of 4. They are also charged because of the two protons that they carry. The relative charge of an alpha particle is +2.

Alpha particles have a short range. The range of ionising radiation is the distance it can travel through matter. Alpha particles can only travel a few centimetres in air and cannot penetrate more than a few millimetres of paper. They have a limited range because they interact with atoms along their paths, causing ions to form. This means that they rapidly give up the energy that they had when they were ejected from the unstable nucleus.

BETA RADIATION (β^- AND β^+)

Beta minus particles (β^-) are very fast-moving electrons that are ejected by a decaying nucleus. The nucleus of an atom contains protons and neutrons, so where does the electron come from? The stability of a nucleus depends on the proportion of protons and neutrons it contains. The result of radioactive decay is to change the balance of protons and neutrons in the nucleus to make it

EXTENSION WORK

Antimatter will not be examined, but for all the particles that exist in ordinary matter there exists an antimatter particle. The positron is the antimatter equivalent of an electron. You will not be asked about neutrinos or antineutrinos either.

more stable. Beta minus decay involves a neutron in the nucleus splitting into a proton and an electron (an antineutrino is also emitted). The proton remains in the nucleus and the electron is ejected at high speed as a beta minus particle.

EXTENSION WORK

Sometimes a proton will decay into a neutron by emitting a beta plus particle (β^+). A beta plus particle is an antielectron, a particle with the same mass as an electron but with positive charge. The antielectron is called a positron.

Beta particles are very light – they have only 0.000 125 times the mass of an alpha particle. The relative charge of a β^- is –1 and the relative charge of a β^+ is +1.

Beta particles interact with matter in their paths less frequently than alpha particles. This is because they are smaller and carry less charge. This means that beta particles have a greater range than alpha particles. Beta particles can travel long distances through air, pass through paper easily and are only absorbed by denser materials like aluminium. A millimetre or two of aluminium foil will stop all but the most energetic beta particles.

GAMMA RAYS (γ)

EXTENSION WORK

Gamma radiation is emitted in 'packets' of energy called photons.

Gamma rays are electromagnetic waves (see page 106) with very short wavelengths. As they are waves, they have no mass and no charge. They are weakly ionising and interact only occasionally with atoms in their paths. They are extremely penetrating and pass through all but the very densest materials with ease. It takes several centimetres thickness of lead, or a metre or so of concrete, to stop gamma radiation.

NEUTRON RADIATION

Neutrons are emitted by radioactive material. They have roughly the same mass as a proton but have no electric charge. The symbol used for a neutron in radioactive decay equations is:

$$^1_0n$$

If a neutron is emitted by a nucleus its mass number (A) goes down by 1 but the atomic number (Z) is unchanged. If a neutron is absorbed by a nucleus its mass number (A) goes up by 1 but the atomic number (Z) is unchanged.

EXTENSION WORK

As neutrons cause other atoms to emit radiation they must be screened. This is done using materials that have a lot of light atoms like hydrogen. Concrete and water are used to prevent neutrons escaping from nuclear reactors.

KEY POINT

As neutrons are not electrically charged they do not directly cause ionisation. They are absorbed by nuclei of other atoms and can cause them to become radioactive. The radioactive nuclei formed in this way will then decay emitting ionising radiation. Neutrons are the only type of radiation to cause other atoms to become radioactive. We shall see later (Chapter 25) that neutron radiation plays an important part in the process of fission used in nuclear reactors.

SUMMARY OF THE PROPERTIES OF IONISING RADIATION

We have said that ionising radiation causes uncharged atoms to lose electrons. An atom that has lost (or gained) electrons has an overall charge. It is called an ion. The three types of radioactive **emission** can all form ions.

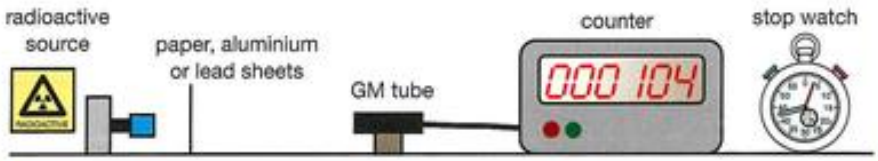
As ionising radiation passes through matter, its energy is absorbed. This means that radiation can only penetrate matter up to a certain thickness. This depends on the type of radiation and the density of the material that it is passing through.

The ionising and penetrating powers of alpha, beta and gamma radiation are compared in the table on page 226. Note that the ranges given in the table are typical but they do depend on the energy of the radiation. For example, more energetic alpha particles will have a greater range than those with lower energy.

Radiation	Ionising power	Penetrating power	Example of range in air	Radiation stopped by
alpha, α	strong	weak	5–8 cm	paper
beta, β	medium	medium	500–1000 cm	thin aluminium
gamma, γ	weak	strong	virtually infinite	thick lead sheet

ACTIVITY 1

▼ PRACTICAL: INVESTIGATE THE PENETRATING POWERS OF DIFFERENT KINDS OF RADIATION



▲ Figure 22.8 Measuring the penetrating power of different types of radiation

Radioactive sources must be stored in lead-lined boxes and kept in a metal cupboard with a radiation warning label. The source must be handled with tongs away from the body. (See Chapter 24 page 247 for more details of safe practice with radioactive materials.)

Before the source is removed from its storage container, measure the background radiation count by connecting a Geiger–Müller (GM) tube to a counter. Write down the number of counts after 5 minutes. Repeat this three times and find the average background radiation count.

Take a source of alpha radiation and set it up at a measured distance (between 2 and 4 cm) from the GM tube.

Measure the counts detected in a 5 minute period. Repeat the count with a sheet of thick paper in front of the source. You should find that the counts have dropped to the background radiation count. This shows that alpha radiation does not pass through paper.

Now replace the alpha source with a beta source. After measuring the new count for 5 minutes place thin sheets of aluminium between the source and detector. When the thickness of the aluminium sheet is 1–2 mm thick you will find that the count has dropped to the background radiation level. This shows that beta radiation is blocked by just a few millimetres of aluminium.

Finally carry out the same steps using a gamma radiation source. Now you will find that gamma radiation is only blocked when a few centimetres of lead are placed between the source and the detector.

! All work with radioactive sources must be closely supervised by qualified science staff and no sources must be touched or pointed towards any part of the body. Source usage must be logged and they must be returned to their secure storage immediately after use. In the UK, students under 16 years are not permitted to handle sources; over 16, they are allowed to do this with close supervision.

KEY POINT

The Geiger–Müller (GM) tube is a detector of radiation. It is described in the next chapter.

As we said earlier, an unstable atom, or strictly speaking its nucleus, will decay by emitting radiation. If the decay process involves the nucleus ejecting either an alpha or a beta particle, the atomic number will change. This means that alpha or beta decay causes the original element to transform into a different element.

ALPHA (α) DECAY

Here is an example of alpha decay:

$$^{222}_{88}\text{Ra} \rightarrow {}^{218}_{86}\text{Ra} + {}^{4}_{2}\text{He} + \text{energy}$$

radium atom → radon atom + α particle + energy

The radioactive isotope radium-222 decays to the element radon by the emission of an alpha particle. The alpha particle is sometimes represented by the Greek letter α. Radon is a radioactive gas that also decays by emitting an alpha particle. Note that the atomic number for radon, 86, is two less than the atomic number for radium.

This is a balanced equation:

- The total of the A numbers on each side of the equation is the same. (Remember that A is the total number of protons and neutrons.)
- The total of the Z numbers on each side of the equation is the same. (Remember that the Z number tells us the number of protons in the nucleus – the number of positively charged particles.)

REMINDER

A is the mass number of the element and Z is the atomic number. The letters W and Y are not the symbols of any particular elements as this is the general equation.

The general form of the alpha decay equation is:

$$^{A}_{Z}\text{Y} \rightarrow {}^{A-4}_{Z-2}\text{W} + {}^{4}_{2}\text{He} + \text{energy}$$

($^{4}_{2}\text{He}$: alpha particle, α)

KEY POINT

It is worth pointing out that the mass number refers to the number of nuclear particles, or nucleons, involved in the transformation – not the exact mass. Mass is not conserved in nuclear transformations, as some of it is transformed into energy.

In alpha decay, element Y is transformed into element W by the emission of an alpha particle. Element W is two places before element Y in the periodic table. The alpha particle, a helium nucleus, carries away four nucleons, which reduces the mass number (A) by four. Two of these nucleons are protons so the atomic number of the new element is two less than the original element, Z – 2. Notice that the mass number and the atomic number are conserved through this equation – that is, the total numbers of nucleons and protons on each side of the equation are the same.

EXAMPLE 1

Complete the following nuclear equation which shows the decay of uranium-238 to the element thorium by alpha decay (emitting an alpha particle).

$$^{238}_{92}\text{U} \rightarrow {}^{?}_{?}\text{Th} \rightarrow {}^{4}_{2}\text{He}$$

The atomic mass numbers (A) on both sides of the equation must add up to the same, so the atomic mass for thorium must be 234 (238 = 234 + 4).

The atomic numbers (Z) on both sides of the equation must add up to the same, so the atomic mass for thorium must be 90 (92 = 90 + 2). So the answer is:

$$^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th} \rightarrow {}^{4}_{2}\text{He}$$

KEY POINT

A complete periodic table is shown in Appendix A on page 277. The numbers shown in the top left corner are the atomic numbers of the elements – the Z numbers in the nuclear equations. Remember that the Z numbers tell us the number of protons in the nucleus of an element. Some periodic tables show the atomic mass too but as elements have a number of different isotopes the mass numbers will be an average for the different isotopes.

Figure 22.9 shows the effect of alpha particle decay. Losing an alpha particle from the nucleus of a uranium atom turns it into thorium, two places back in the periodic table.

Ac	Th	Pa	U	Np
Actinium	Thorium	Protactinium	Uranium	Neptunium
89	90	91	92	93

▲ Figure 22.9 A part of the periodic table of elements

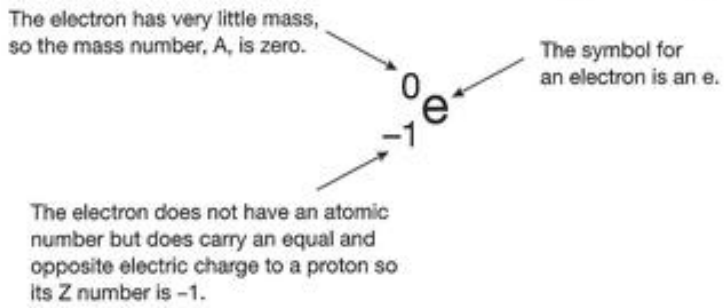
BETA (β^-) DECAY

Here is an example of beta minus decay:

$$^{14}_{6}C \rightarrow {}^{14}_{7}N + {}^{0}_{-1}e + \text{energy}$$

carbon-14 atom → nitrogen-14 atom + beta particle, β^- + energy (also an antineutrino)

The radioactive isotope of carbon, carbon-14, decays to form the stable isotope of the gas nitrogen, by emitting a beta particle. Remember that the beta minus particle is formed when a neutron splits to form a proton and an electron. Figure 22.10 shows the standard atomic notation for a beta particle (β^-).



▲ Figure 22.10 A beta minus particle

The general form of the beta minus decay equation is:

$$^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}e + \text{energy}$$

beta minus particle, β^-

In β^- decay, element X is transformed into element Y by the emission of a beta minus particle. Element Y is the next element in the periodic table after element X. The beta particle, an electron, has practically no mass so the mass number, A, is the same in X and Y. As the β^- particle has a charge of −1, the atomic number of the new element is increased to Z + 1. Again the mass number and the atomic number are conserved through this equation.

BETA (β^+) DECAY

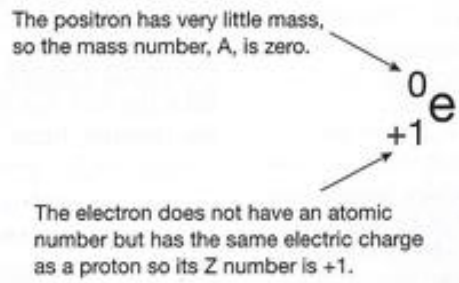
EXTENSION WORK

You will not be asked about β^+ decay but the example here is included to show that the atomic mass numbers (A) and the atomic numbers (Z) on each side of the decay equation must balance. Here is an example of beta plus decay:

$$^{22}_{11}Na \rightarrow {}^{22}_{10}Ne + {}^{0}_{+1}e + \text{energy}$$

sodium-22 atom → neon-22 atom + beta particle, β^+ + energy (also a neutrino)

The radioactive isotope of sodium (Na), sodium-22, decays to form the stable isotope of the gas neon (Ne), by emitting a beta plus particle. Remember that the beta plus particle is formed when a proton splits to form a neutron and a positron. Figure 22.11 shows the standard atomic notation for a beta particle (β^+).



▲ Figure 22.11 A beta plus particle

The general form of the beta plus decay equation is:

$$^{A}_{Z}X \rightarrow {}^{A}_{Z-1}Y + {}^{0}_{+1}e + \text{energy}$$

beta plus particle, β^+

In β^+ decay, element X is transformed into element Y by the emission of a beta plus particle. Element Y is the element in the periodic table that comes before element X. The beta plus particle, a positron, has practically no mass so the mass number, A, is the same in X and Y. As this beta particle has a charge of +1, the atomic number of the new element is decreased to Z – 1.

EXAMPLE 2

An isotope of caesium $^{137}_{55}Cs$ decays by emitting an electron (β^-) and forms an atom of barium (Ba). Write a balanced nuclear equation for this decay.

$$^{137}_{55}Cs \rightarrow {}^{?}_{?}Ba + {}^{0}_{-1}e + \text{energy}$$

An electron has practically no mass so the mass number does not change. The emission of an electron happens when a neutron becomes a proton and an electron, so the atomic number increases by one. The barium isotope formed is therefore $^{137}_{56}Ba$ and the completed equation is:

$$^{137}_{55}Cs \rightarrow {}^{137}_{56}Ba + {}^{0}_{-1}e + \text{energy}$$

When an element decays by emitting an electron it changes to the element with the next higher atomic number. You can check this by looking at the periodic table on page 277.

REMINDER

You will not be asked questions involving positrons in the exam.

EXAMPLE 3

An isotope of magnesium $^{23}_{12}Mg$ decays by emitting a positron (β^+) and forms an atom of sodium (Na). Write a balanced nuclear equation for this decay.

$$^{23}_{12}Mg \rightarrow {}^{?}_{?}Na + {}^{0}_{+1}e$$

A positron has practically no mass so the mass number does not change. The emission of a positron happens when a proton in the nucleus becomes a neutron and a positron, so the atomic number decreases by one. The sodium isotope formed is therefore $^{23}_{11}Na$ and the completed equation is:

$$^{23}_{12}Cs \rightarrow {}^{23}_{11}Na + {}^{0}_{+1}e$$

When an element decays by emitting a positron it changes to the element with the next lower atomic number. You can check this by looking at the periodic table on page 277.

NEUTRON DECAY

Here is an example of neutron decay:

$$^{5}_{2}He \rightarrow {}^{4}_{2}He + {}^{1}_{0}n$$

helium-5 atom → helium-4 atom + neutron

As a neutron has no electric charge, the total positive charge is unchanged by the emission of a neutron – the Z number is 2 before and after the decay. The total number of particles (protons and neutrons) in the nucleus of the helium atom has decreased by 1 because of emission of the neutron.

EXAMPLE 4

An isotope of beryllium, $^{13}_{4}Be$, decays by emitting a neutron. Write a balanced nuclear equation for this transformation.

Write down what you know:

$$^{13}_{4}Be \rightarrow {}^{?}_{?}? + {}^{1}_{0}n$$

The emission of a neutron has no effect on the charge so the atomic number (Z) is unchanged at 4. Each element has its own atomic number so the element in this decay is unchanged – it is still an isotope of beryllium. The loss of a neutron from the beryllium nucleus means that the atomic mass decreases by one. The completed equation is:

$$^{13}_{4}Be \rightarrow {}^{12}_{4}Be + {}^{1}_{0}n$$

GAMMA (γ) DECAY

Gamma radiation is high-energy electromagnetic radiation (see page 106). After an unstable nucleus has emitted an alpha or beta particle it sometimes has surplus energy. It emits this energy as gamma radiation. Gamma rays are pure energy, so they do not have any mass or charge. When a nucleus emits a gamma ray there is no change to either the atomic number or the mass number of the nucleus.

LOOKING AHEAD



▲ Figure 22.12 Inside the Large Hadron Collider

Understanding the particles that make up all matter in the Universe, how they interact and the forces involved holding matter together remains a major area of research for physicists. This is an introduction to what has been called the 'particle zoo'. Large particle accelerators, like the Large Hadron Collider (LHC) on the Swiss/French border, have identified many new particles. Energy turned into matter! Research has recognised patterns in these particles and we now have a model that explains many complex nuclear processes – some of which have increased our understanding of how the Universe was formed.

We also understand how matter can be transformed into energy and the amazing amount of energy that results, revealed by Albert Einstein in his famous equation:

$$E = mc^2$$

CHAPTER QUESTIONS

More questions on the structure of the atom can be found at the end of Unit 7 on page 255.

SKILLS CRITICAL THINKING

7

1 Copy and complete the table below. Identify the particles and complete the missing data.

Atomic particle	Relative mass of particle	Relative charge of particle
	1	−1
		+1
	2000	0

2 Identify the following atomic particles from their descriptions:

a an uncharged nucleon

b the particle with the least mass

c the particle with the same mass as a neutron

d the particle with the same amount of charge as an electron

e a particle that is negatively charged.

3 Explain the following terms used to describe the structure of an atom:

a atomic number

b mass number.

8

4 Copy and complete the table below, describing the structures of the different atoms in terms of numbers of protons, neutrons and electrons.

	$^{3}_{2}He$	$^{13}_{6}C$	$^{23}_{11}N$
protons			
neutrons			
electrons			

5 Copy and complete the following sentences:

a An alpha particle consists of four _____. Two of these are _____ and two are _____. An alpha particle carries a charge of _____.

b A beta minus particle is a fast-moving _____ that is emitted from the nucleus. It is created when a _____ in the nucleus decays to form a _____ and the beta particle.

c A third type of ionising radiation has no mass. It is called _____ radiation. This type of radiation is a type of wave with a very _____ wavelength.

6

d Gamma radiation is part of the _____ spectrum.

SKILLS PROBLEM SOLVING

6 The following nuclear equation showing beta plus decay:

$$^{1}_{1}p \rightarrow {}^{1}_{0}\square + {}^{\square}_{\square}e$$

a From the following list of particles:

A electron B neutron C proton D alpha particle

8

i State which is the particle on the left of this balanced equation.

ii State which is the first of the terms on the right of this equation.

9

b The second term on the right of this equation is a positron – the antiparticle of an electron. Copy and complete the above equation filling in the empty boxes shown.